HanGuk Shin

Product Engineer · Mobile, Media & Distributed Systems

I build complex products across mobile, media, and backend systems, designing them to converge toward the correct state even when operations fail.

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Core Capabilities SCOPE

Product Engineering 0→1 product building · Structuring requirements · App / web / back-office / backend integration · Shipping and operating · Automating a small team's repetitive work

Mobile Android / Kotlin · Jetpack Compose · React Native / Expo · Flutter · Camera · background tasks · App distribution and OTA

Media & Realtime OpenGL ES · EGL · MediaCodec · Video decode / render / encode · WebRTC · STT / TTS · Real-time audio processing

Backend & Distributed Systems Go · TypeScript · Python · PostgreSQL · MQTT · LoRa · Kafka · ClickHouse · State machines · idempotency · Retry · timeout · recovery · observability

Web & Visualization React · Next.js · WebGL · Pannellum · Back-office · Data visualization

Selected Work

REVITA

2025.10 - Present

A computer-vision IoT project where I built the mobile apps, public web, back-office, and field-device control backend from scratch.

ROLE Product engineer - owned the apps (Android/iOS), web, back-office, and field-control backend end to end.

- **15 min+ → ~2 min** Panorama generation time
- Designed the valve-control state machine assuming commands and replies can vanish over LoRa/MQTT - when a reply never comes, timeouts and a safety sweep converge state back to the physical valve
- Designed away dry-running of the pump shared across zones - make-before-break switching and idempotent pump start

React Native · Expo · TypeScript · Next.js · OpenCV · Pannellum · MQTT · LoRa

Android Video Editor

2025.02 - 2025.06

An Android multi-source video editor and encoding pipeline wiring MediaCodec, OpenGL ES, and EGL.

GPU render output wired straight into the encoder, zero CPU copies - Direct EGLSurface hand-off removed the intermediate YUV conversion step.

Android · Kotlin · OpenGL ES · EGL · MediaCodec

Phylaxis Proxy

2023.10 - 2024.01

Built rate limiting, cache, security filtering, and the log/stats pipeline in a Go reverse proxy.

Parsed every request's headers, query, and body to block SQLi/XSS at the proxy

Go · Aerospike · ClickHouse · Kafka · HAProxy

Togispace

2022.01 - 2023.02

Built a commercial Android app from scratch and operated it for 14 months through repeated product pivots.

A Clean Architecture, multi-module design that absorbed repeated product pivots without rewrites (event-driven, reactive)

Kotlin · Jetpack Compose

Avatar Vision

2023.02 - 2023.05

A video-diary app in Flutter and Rust without ffmpeg - decoding webcam/mic input and muxing to MP4.

Fed the fixes back upstream as two PRs to minimp4.rs and nokhwa - Shared the audio-muxing and camera-capture solutions for others hitting the same walls.

Flutter · Dart · Rust · OpenH264 · fdk-aac

Experience ROLES & OWNERSHIP

REVITA · Product Engineer 2025.10 - Present
Yongsan, Seoul

Building and operating the whole product 0→1, from the mobile apps to the field-device control backend.

- Made valve control over unreliable remote devices trustworthy with a state machine and recovery paths
- Cut OpenCV panorama generation from 15+ min to ~2 min, making it usable on-site

Freelance · Android Developer 2025.02 - 2025.06
Project contract

Built an OpenGL/MediaCodec Android video editor.

- Designed and built the decode/compose/encode pipeline
- Wired render frames straight to the encoder via EGLSurface, no CPU copies

Phylaxis · Backend Go Developer 2023.10 - 2024.01
Jung-gu, Seoul

Built cache, rate limit, log, and security features in a Go reverse proxy.

- Rate limit and LRU cache / TTL management
- Input analysis for SQLi/XSS defense

Togispace · Android Developer 2022.01 - 2023.02
Yeouido, Seoul

Sole developer and operator of a commercial B2C app on Kotlin/Jetpack Compose.

- Built a Clean Architecture, multi-module structure
- Kept the app structure and state manageable through repeated pivots

Snaptag · Android & Backend Developer 2021.03 - 2021.06
Pangyo, Gyeonggi

Android camera frame processing and JNI/C++ module integration.

- Optimized from camera capture to information extraction
- Passed camera frames to a C++ module via JNI

Education & Languages

Uiduk University · Information & Communication Engineering (dropped out) 2012.03 - 2018.03

Korea Radio Promotion Association · Cloud training program 2020.05 - 2020.10

Tjoeun Academy · Java web development training 2019.10 - 2020.03

LANGUAGES

EF Language School · English 2017.01 - 2017.06

Meric Japanese School · Japanese 2015.01 - 2015.06

OPEN SOURCE

darkskygit/minimp4.rs#3 · Added audio-track support to MP4 muxing

l1npengtul/nokhwa#105 · Fixed YUV-to-RGB image decoding

Additional Projects

- **VocalBridge** A prototype linking real-time voice through STT, translation, and TTS over React Native and WebRTC.
- **Realtime Chat** Multi-channel real-time messaging with a websocket pool, read receipts, and magic-link login.
- **Strawberry Days** A fan site for singer Younha - aggregated social updates, album/track browsing, and community.
- **Openshy4J** An OpenStack web controller to create and manage instances via Spring and openstack4j.
- **Blue Marble** My first project after learning Java - Monopoly-style board-game logic in the terminal.
- **CrowdFunding** Crowdfunding on a smart contract - campaign creation, funding, and claim, syncing frontend CRUD with on-chain transactions.
- **Plantimer** A productivity app that schedules tasks as a progress-bar timer, continuing via notifications when closed.
- **Snaptag** Scanning invisible QR codes with the Android camera - passing frames to a C++ module via JNI and optimizing capture-to-extraction.
- **Fivestar Recipe** A cooking-recipe site with bookmarks, search, and ranking (Spring MVC, Vue.js).